

Lecture 13

The Heisenberg group, the oscillator, and Fock space

Objectives. Until now every representation in this course has been finite-dimensional by choice. From here on it is infinite-dimensional by theorem: a two-line trace argument shows that the canonical commutation relation $[Q, P] = i\hbar \mathbb{I}$ has no finite-dimensional realisation at all. We package the relation as a Lie algebra and a matrix group — the Heisenberg group, on whose nilpotent algebra (one in which iterated brackets eventually vanish — here after a single step) the Baker–Campbell–Hausdorff series terminates and becomes a theorem of this course; we construct the Schrödinger representation on $L^2(\mathbb{R})$, state Stone–von Neumann uniqueness, and read off why quantum mechanics is one theory; we solve the harmonic oscillator by pure ladder algebra, with coherent states appearing as an orbit of the group; and we let Part I’s symmetric group return to govern many particles — bosonic and fermionic Fock spaces, canonical commutation and anticommutation relations, and second quantisation, with a one-page glance at fields to set up the finale.

13.1 The Heisenberg algebra and group

Quantum kinematics rests on one bracket. Position and momentum of a particle on a line obey

$$[Q, P] = i\hbar \mathbb{I}, \quad (13.1)$$

the *canonical commutation relation*; in this lecture, and only in this one, we keep $\hbar$ visible, because it is about to acquire a group-theoretic meaning. Read as algebra, (13.1) involves three players — Q , P , and the operator $i\hbar \mathbb{I}$ they produce — whose further brackets all vanish. Abstractly: the *Heisenberg algebra* $\mathfrak{h}_3$ is the three-dimensional real Lie algebra with basis X, Y, Z and

$$[X, Y] = Z, \quad [X, Z] = [Y, Z] = 0 :$$

Z is *central*, and it is the whole point, as we shall see. Both the algebra and its group are concrete matrices — the very example Lecture 8’s zoo promised to revisit:

$$\mathfrak{h}_3 = \left\{ \begin{pmatrix} 0 & a & c \\ 0 & 0 & b \\ 0 & 0 & 0 \end{pmatrix} \right\} \subset \mathfrak{gl}(3, \mathbb{R}), \quad H_3 = \left\{ g(a, b, c) = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} \right\} \subset \mathrm{GL}(3, \mathbb{R}),$$

with $X = E_{12}$, $Y = E_{23}$, $Z = E_{13}$ (check the brackets: $E_{12}E_{23} = E_{13}$, the rest vanish). The group H_3 is closed in $\mathrm{GL}(3, \mathbb{R})$ — a matrix Lie group — with multiplication law

$$g(a, b, c) g(a', b', c') = g(a + a', b + b', c + c' + ab') : \quad (13.2)$$

almost a vector space, except for the term ab' , which is where all the physics will live.

Proposition 13.1 (The exponential map, in closed form). For $N \in \mathfrak{h}_3$ with entries (a, b, c) , the series stops: $N^2 = ab E_{13}$, $N^3 = 0$, so

$$e^N = \mathbb{1} + N + \frac{1}{2}N^2 = g(a, b, c + \frac{ab}{2}),$$

and $\exp: \mathfrak{h}_3 \rightarrow H_3$ is a bijection with polynomial inverse.

Better still, on this algebra the Baker–Campbell–Hausdorff series of Lecture 9 — imported there as a black box — terminates and becomes a theorem.

Theorem 13.2 (Exact BCH for step-two nilpotency). Let X, Y be square matrices with $[X, [X, Y]] = [Y, [X, Y]] = 0$. Then

$$e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]}. \quad (13.3)$$

In particular (13.3) holds for all $X, Y \in \mathfrak{h}_3$: the first two terms of Lecture 9's series are the whole series here.

Proof. Set $G(t) = e^{tX} e^{tY}$. Differentiating,

$$G'(t) = XG(t) + e^{tX} Y e^{-tX} G(t) = (X + Y + t[X, Y])G(t),$$

because $e^{tX} Y e^{-tX} = e^{t \operatorname{ad}_X} Y = Y + t[X, Y]$ exactly — the adjoint series of Lecture 9 terminates since $\operatorname{ad}_X^2 Y = 0$. Now set $H(t) = e^{t(X+Y)+\frac{t^2}{2}[X,Y]}$. The two exponents commute ($[X, Y]$ commutes with $X + Y$ by hypothesis), so $H(t) = e^{t(X+Y)} e^{\frac{t^2}{2}[X,Y]}$ and

$$H'(t) = (X + Y)H(t) + t[X, Y]H(t),$$

the central factor sliding through the first exponential. So G and H solve the same linear matrix ODE with the same initial value $\mathbb{1}$; by uniqueness (the ODE course), $G \equiv H$, and $t = 1$ is the claim. ■

Remark 13.3 (The centre is made of phases). By (13.3), two Heisenberg group elements built from X - and Y -directions commute only up to a factor from the centre. In an irreducible unitary representation the central e^{cZ} act as mere *phases* (this needs Schur's lemma for unitary representations on Hilbert spaces — an import, Lecture 4's proof being finite-dimensional; for the Schrödinger representation below it is checked directly) — precisely the currency that Lecture 11 showed quantum mechanics keeps in circulation: states are rays, and phases are free. The Heisenberg centre is thus the abstract home of quantum phases; the irremovable mass-phase of the Galilei group that Lecture 11 mentioned in passing is this centre's close kin, not the same central character: Galilei mass is the eigenvalue of the Galilei central generator, whereas the Heisenberg central character below is labelled by $\hbar$. Translations of position and translations of momentum commute as physical operations; their quantum implementers do not, and the mismatch is a phase — measurable, as interferometry keeps demonstrating.

13.2 No finite-dimensional escape, and the Schrödinger representation

Theorem 13.4 (The trace obstruction). There are no $n \times n$ matrices Q, P (any $n \geq 1$) with $[Q, P] = i\hbar \mathbb{1}$ and $\hbar \neq 0$.

Proof. The trace of a commutator vanishes; the trace of $i\hbar \mathbb{1}$ is $i\hbar n \neq 0$. ■

Two lines, and they redraw the course's map. In any finite-dimensional representation ρ of $\mathfrak{h}_3$, $\text{tr } \rho(Z) = \text{tr}[\rho(X), \rho(Y)] = 0$. If the centre acts by a scalar, $\rho(Z) = c\mathbb{I}$, then $c \dim V = 0$, so $c = 0$. In particular, in an irreducible finite-dimensional complex representation Schur's lemma makes $\rho(Z)$ scalar, and the centre is killed. The scalar hypothesis matters: the defining upper-unitriangular representation on $\mathbb{C}^3$ has $\rho(X) = E_{12}$, $\rho(Y) = E_{23}$, and $\rho(Z) = E_{13} \neq 0$, which is non-scalar. Thus the genuine conclusion is both sharp and strong: a nonzero scalar central character — in particular the canonical commutation relation with $\hbar \neq 0$ — forces infinite dimension. Infinite-dimensionality of quantum mechanics is not mathematical extravagance; it is a theorem, and the rest of the lecture lives with its consequences.

Remark 13.5 (Scope and operator-theory boundary). From here the representation is a strongly continuous unitary representation on a separable Hilbert space (one with a countable orthonormal basis). Its bounded group operators are honest everywhere-defined unitaries; its derived generators are generally unbounded and all algebraic identities involving $Q, P, a, a^\dagger$ are first statements on the named common invariant dense domain. Essential self-adjointness, closures, spectral exponentials, and extension from the core are point-of-use imports. Finite-dimensional trace arguments above and finite-sector calculations below remain valid only in their stated scopes.

The representation that exists is the one Schrödinger wrote down. On $L^2(\mathbb{R})$ set

$$(Q\psi)(x) = x\psi(x), \quad (P\psi)(x) = -i\hbar\psi'(x).$$

Both operators are unbounded — no escape, by two explicit tests, not by the finite-dimensional trace argument. Choose $\phi \in C_c^\infty(0, 1)$ with $\|\phi\| = 1$ and put $\phi_n(x) = \phi(x - n)$. Every ϕ_n is smooth with compact support (in particular it lies in the Schwartz space introduced below), its norm is one, and its support lies in $(n, n+1)$, so $\|Q\phi_n\| \geq n$. For momentum choose a fixed nonzero $\phi \in \mathcal{S}(\mathbb{R})$, normalised to norm one, and put $\psi_n(x) = e^{inx}\phi(x)$. Again $\psi_n \in \mathcal{S}(\mathbb{R})$ and $\|\psi_n\| = 1$, while our sign convention gives

$$P\psi_n = \hbar n\psi_n + e^{inx}P\phi, \quad \|P\psi_n\| \geq \hbar n - \|P\phi\| \longrightarrow \infty.$$

Thus multiplication by x and $-i\hbar d/dx$ are each unbounded on the stated domain.

There is also a distinct infinite-dimensional obstruction, due to Wintner: no two bounded operators on a complex Hilbert space (indeed, on a Banach space) satisfy $[Q, P] = i\hbar\mathbb{I}$ with $\hbar \neq 0$. For if they did, induction would give $[Q, P^n] = n i\hbar P^{n-1}$. No power P^{n-1} can vanish: if $P^k = 0 \neq P^{k-1}$ then $0 = [Q, P^k] = k i\hbar P^{k-1}$, a contradiction. Hence

$$n|\hbar| \|P^{n-1}\| \leq 2\|Q\| \|P^n\| \leq 2\|Q\| \|P\| \|P^{n-1}\|.$$

Cancellation says $n|\hbar| \leq 2\|Q\| \|P\|$ for every n , a contradiction. This norm argument, not the trace proof, excludes bounded CCR pairs in infinite dimension (Reed–Simon I, Section VIII.5, Example 2, p. 275).

The honest setting our functional analysis supports is the Schwartz space $\mathcal{S}(\mathbb{R})$ of rapidly decreasing smooth functions, a dense subspace invariant under both operators, on which (13.1) is the product rule: $(QP - PQ)\psi = -i\hbar(x\psi' - (x\psi)') = i\hbar\psi$. That Q and P are essentially self-adjoint on this domain, and that their closures have the stated spectral exponentials, follows from the invariant-domain generator criterion in Reed–Simon I, Theorem VIII.10, applied to the modulation and translation groups below (with Stone's theorem VIII.8). Students use the result; they are not expected to prove the domain and closure assertions here.

The genuinely group-theoretic objects are the bounded Weyl unitaries, where the domain work has been packaged into the imported spectral calculus. By Stone's theorem read backwards through Lecture 9's flagship examples,

$$(U(a)\psi)(x) = \psi(x - a) \text{ has generator } P, \quad (V(b)\psi)(x) = e^{ibx/\hbar}\psi(x) \text{ has generator } Q,$$

that is, $U(a) = e^{-iaP/\hbar}$ and $V(b) = e^{ibQ/\hbar}$ — honest unitaries (“generator” here means the operator in the exponent up to these conventional factors; Stone's generator in the normalisation e^{-itA} of Lecture 9 is $P/\hbar$, respectively $-Q/\hbar$). They satisfy the *Weyl relation*: for all $a, b \in \mathbb{R}$,

$$U(a)V(b) = e^{-iab/\hbar} V(b)U(a), \quad (13.4)$$

verified in one line each way: $(U(a)V(b)\psi)(x) = e^{ib(x-a)/\hbar}\psi(x-a)$ while $(V(b)U(a)\psi)(x) = e^{ibx/\hbar}\psi(x-a)$. Equation (13.4) is the commutation relation (13.1) with the calculus removed: bounded operators, exact identity, and the obstruction to commuting is a pure phase — the centre of Remark 13.3 arriving on stage.

This is exactly the limited quantisation debt recorded in Lecture 9's Noether dictionary: Theorem 13.4 gives the finite-dimensional trace obstruction and the bounded-CCR obstruction, Q and P are instead unbounded on their common Schwartz core, and (13.4) is the bounded Weyl form. No general quantisation map for arbitrary classical observables is asserted here.

Proposition 13.6 (The Schrödinger representation). *The assignment*

$$\Pi(g(a, b, c)) = e^{-ic/\hbar} V(b)U(a)$$

is a strongly continuous unitary representation of H_3 on $L^2(\mathbb{R})$, in which the central subgroup $g(0, 0, c)$ acts by the phases $e^{-ic/\hbar}$. It is irreducible.

Proof. Multiply and use (13.4):

$$\Pi(a, b, c) \Pi(a', b', c') = e^{-i(c+c')/\hbar} V(b)U(a)V(b')U(a') = e^{-i(c+c'+ab')/\hbar} V(b+b')U(a+a'),$$

which is Π of the product (13.2). Strong continuity: for translations this is Lecture 9's density argument; for the multiplications it is dominated convergence (an analysis-course theorem used without proof). Irreducibility is Sheet 13's guided exercise; the idea in one breath: a projection commuting with all $V(b)$ commutes with multiplication by every function, hence is itself a multiplication operator, and commuting also with all translations $U(a)$ forces it to be constant — 0 or $\mathbb{1}$. ■

Planck's constant has become a label on the centre: the representation Π has central character $c \mapsto e^{-ic/\hbar}$, and representations with different $\hbar$ are inequivalent, since unitary equivalence preserves the central phases. Within one value of $\hbar$, however, there is nothing to choose — the central import of this lecture:

Theorem 13.7 (Stone–von Neumann; imported). *Every strongly continuous irreducible unitary representation of H_3 on a separable Hilbert space whose central subgroup acts by the phases $e^{-ic/\hbar}$, for a fixed $\hbar \neq 0$, is unitarily equivalent to the Schrödinger representation.*

Reference for the omitted proof: Folland, *Harmonic Analysis in Phase Space*, Theorem 1.50; Woit, the Heisenberg-group chapters. The full Folland entry appears in the consolidated bibliography.

This is the uniqueness theorem behind the sentence “quantum mechanics is one theory”. In 1925–26 Heisenberg’s matrix mechanics and Schrödinger’s wave mechanics computed the same spectra by what looked like different physics; Theorem 13.7 says that *any* two realisations of the same canonical commutation relations are the same theory in different unitary coordinates. (Starred, and ominous: the theorem *fails* for infinitely many degrees of freedom. Quantum field theory lives with inequivalent representations of its commutation relations — inequivalent vacua — and Section 13.5 will peek through that door.)

13.3 The harmonic oscillator

The oscillator, $H = \frac{P^2}{2m} + \frac{m\omega^2 Q^2}{2}$, is the most-solved problem in physics; we solve it once more because the solution is pure representation theory. Define

$$a = \frac{m\omega Q + iP}{\sqrt{2m\omega\hbar}}, \quad a^\dagger = \frac{m\omega Q - iP}{\sqrt{2m\omega\hbar}}, \quad [a, a^\dagger] = \mathbb{1}, \quad (13.5)$$

the bracket from (13.1), and with $N = a^\dagger a$ a direct expansion gives

$$H = \hbar\omega\left(N + \frac{1}{2}\right), \quad [N, a] = -a, \quad [N, a^\dagger] = a^\dagger. \quad (13.6)$$

The last two brackets are Lecture 10’s ladder relations with $(N; a^\dagger, a)$ in the roles of $(J_z; J_+, J_-)$ — but with one structural change: the bracket of the ladders, $[a, a^\dagger] = \mathbb{1}$, is now *central*. The staircase logic survives; its termination does not.

Theorem 13.8 (The oscillator spectrum, by algebra alone). *Let ψ be a normalised eigenvector of N in the invariant domain $\mathcal{S}(\mathbb{R})$, $N\psi = \lambda\psi$. Then $\lambda \in \mathbb{Z}_{\geq 0}$; the vector $a^\lambda\psi$ is a nonzero multiple of a vector ψ_0 with $a\psi_0 = 0$; and from any such normalised ψ_0 the vectors*

$$\psi_n = \frac{(a^\dagger)^n}{\sqrt{n!}} \psi_0, \quad N\psi_n = n\psi_n, \quad a^\dagger\psi_n = \sqrt{n+1}\psi_{n+1}, \quad a\psi_n = \sqrt{n}\psi_{n-1},$$

are orthonormal. Hence the energies are $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$, $n = 0, 1, 2, \dots$

Proof. Norms, not hand-waving. First, $\lambda = \langle \psi, a^\dagger a \psi \rangle = \|a\psi\|^2 \geq 0$. From $[N, a] = -a$, the vector $\varphi_k = a^k\psi$ obeys $N\varphi_k = (\lambda - k)\varphi_k$, and the recursion

$$\|\varphi_{k+1}\|^2 = \langle \varphi_k, a^\dagger a \varphi_k \rangle = (\lambda - k) \|\varphi_k\|^2$$

gives $\|\varphi_k\|^2 = \lambda(\lambda - 1) \cdots (\lambda - k + 1)$. If λ were not a non-negative integer, no factor would vanish and the product would eventually turn negative — a negative squared norm. So $\lambda = n \in \mathbb{Z}_{\geq 0}$, the product stops exactly at $\|\varphi_{n+1}\|^2 = 0$, and $\varphi_n \neq 0$ ($\|\varphi_n\|^2 = n!$) is annihilated by a . Climbing back up, $\|a^\dagger\varphi\|^2 = \langle \varphi, (N + \mathbb{1})\varphi \rangle = (\mu + 1) \|\varphi\|^2$ on an eigenvector of eigenvalue μ : never zero, so the upward ladder never ends, and the stated normalisations follow from the same two norm formulas. Eigenvectors of distinct eigenvalues are orthogonal. ■

No differential equation was harmed: the spectrum $\hbar\omega\left(n + \frac{1}{2}\right)$ — zero-point energy included — is bracket algebra plus positivity of the norm. Analysis enters only to ground the staircase. In the Schrödinger representation, $a\psi_0 = 0$ is the first-order ODE $(m\omega x + \hbar \frac{d}{dx})\psi_0 = 0$, whose solution space is one-dimensional:

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/2\hbar}, \quad \psi_n(x) \propto H_n(\xi) e^{-\xi^2/2}, \quad \xi = \sqrt{\frac{m\omega}{\hbar}} x,$$

the Hermite functions, H_n being the *Hermite polynomials*: the ladder produces, at each step, a polynomial of degree n with nonzero leading coefficient times the Gaussian, so the span of the ψ_n is the span of the $\xi^n e^{-\xi^2/2}$. They are complete in $L^2(\mathbb{R})$ — a point-of-use import, stated without proof: the span of the functions $\xi^n e^{-\xi^2/2}$ is dense in $L^2(\mathbb{R})$ — so the eigenvalues above exhaust the spectrum, and $L^2(\mathbb{R})$ is exactly the closure of the staircase: the Schrödinger representation is the oscillator's ladder, nothing more.

Proposition 13.9 (Coherent states: the orbit of the vacuum). *For $\alpha \in \mathbb{C}$ put*

$$q_\alpha = \sqrt{\frac{2\hbar}{m\omega}} \Re \alpha, \quad p_\alpha = \sqrt{2m\omega\hbar} \Im \alpha,$$

and define, first and globally, by bounded Weyl unitaries

$$D(\alpha) := e^{iq_\alpha p_\alpha / 2\hbar} U(q_\alpha) V(p_\alpha). \quad (13.7)$$

Then $D(\alpha)$ is unitary,

$$D(\alpha)D(\beta) = e^{(\alpha\bar{\beta} - \bar{\alpha}\beta)/2} D(\alpha + \beta), \quad D(\alpha)^* = D(\alpha)^{-1} = D(-\alpha).$$

If $|\alpha\rangle := D(\alpha)\psi_0$, then

$$a|\alpha\rangle = \alpha|\alpha\rangle, \quad |\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n \geq 0} \frac{\alpha^n}{\sqrt{n!}} \psi_n.$$

On $\mathcal{S}(\mathbb{R})$, $A_\alpha := \alpha a^\dagger - \bar{\alpha}a = ip_\alpha Q/\hbar - iq_\alpha P/\hbar$ is essentially skew-adjoint and $D(\alpha) = e^{\bar{A}_\alpha}$. This last statement is a consequence of the invariant-domain generator theorem, not of formal anti-Hermiticity. Thus coherent states are the orbit of the vacuum under H_3 — not a trick, an orbit.

Figure 13.1 is the proposition as a picture: the vacuum is a minimum-uncertainty blob at the origin, $|\alpha\rangle$ is that blob displaced by α , and harmonic evolution $\alpha \mapsto \alpha e^{-i\omega t}$ slides it rigidly around a circle — the classical orbit, with the quantum spread carried along unchanged.

Proof. The constants are forced by (13.5):

$$\alpha a^\dagger - \bar{\alpha}a = \frac{2im\omega(\Im \alpha)Q - 2i(\Re \alpha)P}{\sqrt{2m\omega\hbar}} = \frac{ip_\alpha Q - iq_\alpha P}{\hbar}.$$

The phase in (13.7) is checked without unbounded calculus. For every $\psi \in \mathcal{S}(\mathbb{R})$,

$$(D(\alpha)\psi)(x) = e^{ip_\alpha(x - q_\alpha/2)/\hbar} \psi(x - q_\alpha). \quad (13.8)$$

This proves unitarity directly. Multiplying two such formulas gives the phase identity

$$e^{i(q_\beta p_\alpha - q_\alpha p_\beta)/2\hbar} = e^{(\alpha\bar{\beta} - \bar{\alpha}\beta)/2},$$

and also gives the adjoint formula. Moreover

$$D(\alpha) = e^{-iq_\alpha p_\alpha / 2\hbar} V(p_\alpha) U(q_\alpha) = \Pi(g(q_\alpha, p_\alpha, q_\alpha p_\alpha / 2)),$$

so the same phase follows from the Heisenberg law (13.2).

For fixed α , (13.8) shows that $t \mapsto D(t\alpha)$ is a strongly continuous one-parameter unitary group, preserves $\mathcal{S}(\mathbb{R})$, and has derivative A_α there. Reed–Simon I, Theorem VIII.10 therefore

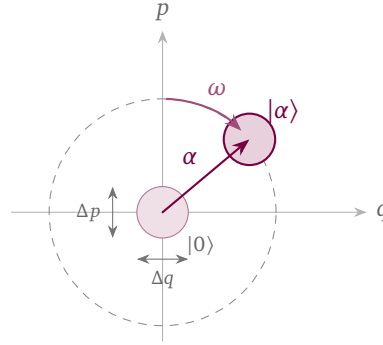


Figure 13.1: A coherent state in the phase plane. The vacuum $|0\rangle$ is the minimum-uncertainty blob at the origin, $\Delta q \Delta p = \hbar/2$; the coherent state $|\alpha\rangle$ is that same blob rigidly displaced by α , with $(\langle q \rangle, \langle p \rangle) \propto (\Re \alpha, \Im \alpha)$. Under the harmonic Hamiltonian it evolves by $\alpha \mapsto \alpha e^{-i\omega t}$: the blob rides the dashed circle at angular frequency ω without spreading, tracing the classical orbit. Coherent states are the orbit of the vacuum under the Heisenberg group, the closest a quantum state comes to a classical point in phase space. ($\hbar$ is kept explicit throughout this lecture.)

says that A_α is essentially skew-adjoint and that its closure generates this group; at $t = 1$ this is $D(\alpha) = e^{\bar{A}\alpha}$.

The normal-ordering formula is used only as a vector identity on the analytic finite-particle core $\mathcal{D}_{\text{fin}} = \text{span}\{\psi_n : n \geq 0\}$:

$$D(\alpha)f = e^{-|\alpha|^2/2} e^{\alpha a^\dagger} e^{-\bar{\alpha}a} f, \quad f \in \mathcal{D}_{\text{fin}}. \quad (13.9)$$

Here the annihilation series terminates and the creation series converges absolutely, since $\|(a^\dagger)^r \psi_n\| = \sqrt{(n+r)!/n!}$; commuting the two absolutely convergent series with $[a, a^\dagger] = \mathbb{K}$ proves (13.9). This is the analytic-vector BCH argument (Reed–Simon I, Section VIII.5, pp. 275–276), not a global finite-matrix BCH claim about closed unbounded operators.

Applying (13.9) to ψ_0 gives the displayed series. It is a unit vector because the squared coefficients sum to $e^{-|\alpha|^2} e^{|\alpha|^2} = 1$. It also belongs to the domain of the closed annihilation operator, since

$$\sum_{n \geq 0} n e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!} = |\alpha|^2 < \infty.$$

The ladder formula may therefore be applied term by term and gives $a|\alpha\rangle = \alpha|\alpha\rangle$. Minimal uncertainty and the classical motion of $\langle Q \rangle, \langle P \rangle$ are Sheet 13's. ■

Remark 13.10 (Bargmann–Fock, starred). Lecture 10 promised that its Gaussian inner product would be industrialised; here is the factory. On entire functions $f(z)$ with $\frac{1}{\pi} \int_{\mathbb{C}} |f|^2 e^{-|z|^2} dA < \infty$ — Lecture 10's inner product with the degree restriction removed — set $a^\dagger = z$ (multiplication) and $a = d/dz$. On the polynomial core they are adjoint: integrate $\partial_{\bar{z}}(\bar{f} g e^{-|z|^2})$ over expanding discs; its boundary term tends to zero by the Gaussian, leaving $\langle f', g \rangle = \langle f, zg \rangle$. The Hilbert-space completion and the Bargmann transform are starred further reading. Thus $[a, a^\dagger] = \mathbb{K}$ by the product rule. The vacuum is the constant 1, the staircase is $z^n/\sqrt{n!}$, and the whole oscillator becomes complex analysis. It is the exact analytic sibling of the polynomial model: there the central bracket was $2J_z$ and the staircase stopped at degree $2j$; here the bracket is central and the staircase climbs forever — which

is why the Hilbert space had to become infinite-dimensional spaces of entire functions. By Stone–von Neumann this model is unitarily equivalent to Schrödinger’s.

13.4 Many particles: the symmetric group returns

One particle has a Hilbert space $\mathcal{H}$; n particles of the same kind have $\mathcal{H}^{\otimes n}$ (for the finite-dimensional $\mathcal{H} = \mathbb{C}^d$ used from Definition 13.11 on, the ordinary tensor power; in general, its Hilbert-space completion), and on it two groups act at once. The symmetric group S_n — Part I’s protagonist — permutes the factors,

$$P_\sigma(v_1 \otimes \cdots \otimes v_n) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(n)},$$

and any one-particle symmetry acts diagonally, $U \mapsto U^{\otimes n}$. The two actions commute — permuting identical copies of U does nothing — and that innocuous sentence is the skeleton of many-body physics. Inside $\mathcal{H}^{\otimes n}$ sit the *symmetric* and *antisymmetric* subspaces,

$$\text{Sym}^n \mathcal{H} = \{\psi : P_\sigma \psi = \psi \ \forall \sigma\}, \quad \Lambda^n \mathcal{H} = \{\psi : P_\sigma \psi = \text{sgn}(\sigma) \psi \ \forall \sigma\},$$

the isotypic components of the two one-dimensional representations of S_n , both invariant under every $U^{\otimes n}$ because the actions commute.

Now the physics. For *identical* particles, permutations relabel our bookkeeping, so physical observables are permutation-invariant and the state space splits into S_n representation sectors. Ray invariance alone does not force every ray to carry a one-dimensional character: higher-dimensional irreducible sectors are also allowed and are called *parastatistics*. Permutation type is a superselection label under permutation-invariant observables (no such observable connects different types); that statement must not be confused with selecting only two types.

For ordinary nonrelativistic quantum mechanics we therefore add the *symmetrisation postulate* as physical input: a species is assigned either the trivial S_n sector $\text{Sym}^n \mathcal{H}$ (*bosons*) or the sign sector $\Lambda^n \mathcal{H}$ (*fermions*). This is the adopted input for the Fock-space construction below, not a consequence of ray kinematics. The relativistic spin–statistics theorem, which relates spin to that assignment under additional physical hypotheses, is a distinct result outside this course’s proof scope. (In two space dimensions exchange is braided rather than permutational, allowing anyons; starred reading.) Messiah–Greenberg is the source for the independent nonrelativistic symmetrisation postulate and its experimental foundation; it is not being cited as a proof of relativistic spin–statistics. For $n = 2$, $\mathcal{H} = \mathbb{C}^2$, this is Lecture 12’s singlet–triplet split: the duet, now a chorus. Starred, for perspective: the *full* decomposition of $(\mathbb{C}^d)^{\otimes n}$ under the commuting pair $S_n \times \text{U}(d)$ pairs their irreducible representations one-to-one through a single combinatorics — Schur–Weyl duality, the proper sequel to Part I.

Letting n vary gives the arena of variable particle number.

Definition 13.11 (Fock spaces over $\mathbb{C}^d$). For a one-particle space $\mathbb{C}^d$ (“ d modes”), the bosonic and fermionic Fock spaces are

$$\mathcal{F}_+ = \bigoplus_{n \geq 0} \text{Sym}^n \mathbb{C}^d, \quad \mathcal{F}_- = \bigoplus_{n=0}^d \Lambda^n \mathbb{C}^d,$$

with the $n = 0$ summand $\mathbb{C}$ spanned by the *vacuum* Ω . Orthonormal bases are the occupation-number states $|n_1, \dots, n_d\rangle$ — n_i particles in mode i — with $n_i \in \mathbb{Z}_{\geq 0}$ for bosons and $n_i \in \{0, 1\}$

for fermions (antisymmetry kills repeated factors: the *Pauli exclusion principle* is the sign character at work). So $\mathcal{F}_+$ is infinite-dimensional and $\dim \mathcal{F}_- = 2^d$ is finite. The algebraic finite-particle cores are

$$\mathcal{F}_{+, \text{alg}} = \bigoplus_{n \geq 0}^{\text{alg}} \text{Sym}^n \mathbb{C}^d, \quad \mathcal{F}_{-, \text{alg}} = \bigoplus_{n=0}^d \Lambda^n \mathbb{C}^d = \mathcal{F}_-;$$

we write $\mathcal{F}_{\text{alg}}$ when the choice of statistics is clear.

On $\mathcal{F}_+$, define for each mode the creation and annihilation operators by the one-oscillator formulas of Theorem 13.8, run in parallel:

$$a_i^\dagger |\dots, n_i, \dots\rangle = \sqrt{n_i + 1} |\dots, n_i + 1, \dots\rangle, \quad a_i |\dots, n_i, \dots\rangle = \sqrt{n_i} |\dots, n_i - 1, \dots\rangle,$$

whence, checked mode by mode on the basis,

$$[a_i, a_j^\dagger] = \delta_{ij} \mathbb{K}, \quad [a_i, a_j] = [a_i^\dagger, a_j^\dagger] = 0 : \quad (13.10)$$

the *CCR algebra* of d modes — d commuting copies of the oscillator staircase, woven into one space. On $\mathcal{F}_-$ the analogues carry a sign string: with $s_i = \sum_{j < i} n_j$,

$$c_i^\dagger |\dots, n_i, \dots\rangle = (-1)^{s_i} \delta_{n_i, 0} |\dots, 1, \dots\rangle, \quad c_i |\dots, n_i, \dots\rangle = (-1)^{s_i} \delta_{n_i, 1} |\dots, 0, \dots\rangle,$$

and the brackets become *anticommutators*:

$$\{c_i, c_j^\dagger\} = \delta_{ij} \mathbb{K}, \quad \{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0, \quad (13.11)$$

the *CAR algebra*. For $i = j$ the first relation is a two-line basis check (one of $c_i c_i^\dagger$, $c_i^\dagger c_i$ acts as the identity on each basis vector, the other as zero); for $i \neq j$ the string is the whole point — exchanging the order of two operations shortens exactly one sign string by one, flipping the sign — and Sheet 13 writes out $d = 2$ in full.

Proposition 13.12 (Fermions slip through the no-go). *The CAR algebra (13.11) of d modes has the faithful 2^d -dimensional representation above: finite, in pointed contrast to Theorem 13.4. There is no contradiction: the trace of a commutator vanishes identically, while the trace of an anticommutator is $2 \text{tr}(c_i c_i^\dagger)$, perfectly capable of equalling $\text{tr} \mathbb{K}$.*

The operators $\gamma_{2i-1} = c_i + c_i^\dagger$ and $\gamma_{2i} = i(c_i - c_i^\dagger)$ are Hermitian and satisfy $\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu} \mathbb{K}$: a *Clifford algebra*. Lecture 14's γ -matrices and spinors are exactly this structure with the Euclidean δ replaced by the metric of spacetime — the fermionic oscillator is the spinor's nursery.

Finally, symmetries. A one-particle observable or generator $A \in \mathfrak{gl}(\mathbb{C}^d)$ should act on each particle in turn; Lecture 12's product rule, iterated, makes this precise.

On each fixed n -particle sector the following operators are ordinary finite matrices. On the full bosonic Fock direct sum they are generally unbounded; their algebraic identities are asserted on the algebraic finite-particle core $\mathcal{F}_{\text{alg}}$, which every displayed operator preserves. Closure and exponentiation from that core are imported operator theory, not a consequence of the finite-sector calculation.

Definition 13.13 (Second quantisation). For $A \in \mathfrak{gl}(\mathbb{C}^d)$, define $d\Gamma(A)$ on the n -particle sector of $\mathcal{F}_{\pm, \text{alg}}$ by

$$d\Gamma(A)|_n = \sum_{k=1}^n \mathbb{K} \otimes \dots \otimes A_{k\text{-th}} \otimes \dots \otimes \mathbb{K}, \quad (13.12)$$

which commutes with every P_σ and therefore preserves Sym^n and Λ^n .

Theorem 13.14 (Symmetries lift to Fock space). $d\Gamma$ is a Lie algebra homomorphism on $\mathcal{F}_{\text{alg}}$:

$$[d\Gamma(A), d\Gamma(B)] = d\Gamma([A, B]).$$

Every operator in this identity preserves that core. At group level, for a unitary one-particle operator U , $\Gamma(U) = \bigoplus_n U^{\otimes n}$ (restricted to $\mathcal{F}_{\pm}$) is an everywhere-defined unitary. If A is anti-Hermitian, then $d\Gamma(A)$ is essentially skew-adjoint on $\mathcal{F}_{\text{alg}}$ and

$$\Gamma(e^{tA}) = e^{t \overline{d\Gamma(A)}}.$$

Equivalently, for a Hermitian one-particle observable H , $d\Gamma(H)$ is essentially self-adjoint there and $\Gamma(e^{-itH}) = e^{-it \overline{d\Gamma(H)}}$. The core/closure assertion is the imported second-quantisation theorem (Bratteli–Robinson, Vol. 2, Section 5.2.1, pp. 7–8); the derivative on $\mathcal{F}_{\text{alg}}$ is (13.12) by Lecture 12’s product rule on each fixed sector. No unitary all-Fock conclusion is asserted for arbitrary $A \in \mathfrak{gl}(\mathbb{C}^d)$.

Proof. On the n -sector, writing A_k for A in slot k : $[\sum_k A_k, \sum_l B_l] = \sum_{k,l} [A_k, B_l]$, and operators in different slots commute, so only $k = l$ survives: $\sum_k [A, B]_k = d\Gamma([A, B])|_n$. ■

In mode language, $d\Gamma(A) = \sum_{ij} A_{ij} a_i^\dagger a_j$ (and the same with c ’s) — Sheet 13 takes this formula as the definition on the finite-particle core and proves it is a Lie-algebra homomorphism; it agrees with (13.12) because both sides kill Ω and satisfy $[\cdot, a_k^\dagger] = a_{Ae_k}^\dagger$ — with the flagship special case $N = d\Gamma(\mathbb{1}) = \sum_i a_i^\dagger a_i$, initially on $\mathcal{F}_{\text{alg}}$; as usual N denotes its self-adjoint closure. On the bosonic occupation vector with total particle number n it has eigenvalue n , so N is unbounded on $\mathcal{F}_+$. Second quantisation is thus a machine that eats one-particle symmetries and outputs many-body symmetries, brackets intact; when Lecture 15 needs the Poincaré group to act on multi-particle states, this is the machine that will be running.

13.5 Outlook: fields

One page of honesty about where this leads. Put a free scalar particle on a ring of length L : the one-particle space has modes e^{ikx} with $k \in \frac{2\pi}{L}\mathbb{Z}$, and a free *field* is, by definition, the bosonic Fock construction applied to all of them at once, with Hamiltonian

$$H = \sum_k \hbar \omega_k (a_k^\dagger a_k + \frac{1}{2}) :$$

a field is countably many oscillators, with frequencies $\omega_k = |k|$ for a massless field ($\sqrt{k^2 + m^2}$ with mass m). For any finite set of modes, everything in this lecture is rigorous and nothing new happens. The new phenomena are exactly the infinities. First, the zero-point sum $\sum_k \hbar \omega_k / 2$ diverges — the first renormalisation: energies are measured from the vacuum, so one subtracts it (“normal ordering” in the field-theory sense of moving every $a^\dagger$ to the left of every a , which removes exactly this constant — a different use of the phrase from the normal-ordering formula (13.9)), though its *variation* with boundary conditions survives as measurable physics, the Casimir effect. Second, and deeper, Stone–von Neumann fails: for infinitely many modes the CCR algebra has *inequivalent* irreducible representations (not on the finite ring above — there a change of mass is still implemented by a unitary, the mode-by-mode mixing coefficients being square-summable — but in infinite volume), and physically distinct situations — different masses, different phases of matter, interacting versus free vacua — inhabit different ones. The uniqueness that made quantum mechanics

one theory makes quantum field theory many; taming that plurality is a different course. What this course can still settle is the question the construction begs: a relativistic field should carry a representation of *which* group? Of the symmetries of spacetime — and so the finale begins: the Lorentz group (Lecture 14) and the Poincaré group (Lecture 15), where the infinite-dimensional unitary representations this lecture has normalised stop being optional.

Remark 13.15 (Computer algebra). Notebook 13 truncates $a, a^\dagger$ to $N_{\max}$ -dimensional matrices and watches the spectrum of $\hbar\omega(a^\dagger a + \frac{1}{2})$ converge as the cutoff grows; animates a coherent state's phase-space ellipse following the classical trajectory; builds the fermionic $c_i, c_i^\dagger$ as Kronecker products of Pauli matrices (Jordan–Wigner in miniature) and verifies (13.11) exactly; and checks the homomorphism $[d\Gamma(A), d\Gamma(B)] = d\Gamma([A, B])$ on random matrices to machine precision.

Summary.

- *No-go.* The canonical relation $[Q, P] = i\hbar\mathbb{1}$ has no finite-dimensional representation (the trace obstruction, Theorem 13.4).
- *Heisenberg group.* Step-two nilpotent, so Baker–Campbell–Hausdorff terminates (Proposition 13.1, Theorem 13.2), giving the Weyl form of the relations (eq. (13.4)).
- *Uniqueness.* The Schrödinger representation on $L^2(\mathbb{R})$ (Proposition 13.6) is the unique irreducible one (Stone–von Neumann, Theorem 13.7).
- *Oscillator and coherent states.* Ladder operators fix the spectrum by algebra alone (Theorem 13.8); coherent states are the orbit of the vacuum under the Heisenberg group (Proposition 13.9).
- *Fock space.* Bosonic and fermionic Fock spaces realise the CCR and CAR (Definition 13.11, Proposition 13.12); symmetries lift by second quantisation $d\Gamma$ (Theorem 13.14).

Tutorial. Exercise Sheet 13.

References

Reed–Simon I, Chapter VIII, especially Theorems VIII.8 and VIII.10 and Section VIII.5, Example 2; Folland, Chapter 1, Sections 2–5 and Theorem 1.50; Woit Chs. 13, 22–23 (Heisenberg group, oscillator, Bargmann–Fock model, and coherent states), with the chapters on fermionic oscillators and Fock space; Bratteli–Robinson, Vol. 2, Section 5.2.1, pp. 7–8 (second quantisation); Hall (the nilpotent case of BCH); Osborn's notes; Gilmore (Heisenberg-group examples); Messiah–Greenberg, *Physical Review* **136** (1964), B248–B267 (nonrelativistic symmetrisation postulate and experimental foundation).

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